BDEDA
CBBDD
DCEBD
DDBAC
B

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1. A city transit authority increases the price of sub- way and bus tickets from $1.25 to $1.50. If the demand for these tickets is price inelastic, the number of people riding buses and subways and the city's revenues will most likely change in which of the following ways?

(A)

Number of People Riding	City's Revenues
Increase	Increase

(B) ✓

Number of People Riding	City's Revenues
Decrease	Increase

(C)

Number of People Riding	City's Revenues
Decrease	Decrease

(D)

Number of People Riding	City's Revenues
Decrease	Remain constant

(E)

Number of People Riding	City's Revenues
Remain constant	Increase

2. A monopolist produces two unrelated goods, X and Y. The demand for X is currently price elastic and the demand for Y is currently price inelastic. To increase its total revenue, the firm should change the price of X and Y in which of the following ways?

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(A)

Price of X	Price of Y
Increase	No change

(B)

Price of X	Price of Y
Increase	Decrease

(C)

Price of X	Price of Y
Increase	Increase

(D) ✓

Price of X	Price of Y
Decrease	Increase

(E)

Price of X	Price of Y
Decrease	Decrease

3. Assume a consumer finds that his total expenditure on compact discs stays the same after the price of compact discs declines. Which of the following is true for this price change?

(A) Compact discs are inferior goods to this consumer.
(B) The consumer's demand for compact discs increased in response to the price change.
(C) The consumer's demand for compact discs is perfectly price elastic.
(D) The consumer's demand for compact discs is perfectly price inelastic.
(E) The consumer's demand for compact discs is unit price elastic. ✓

4. Assume that a monopolist is producing in the inelastic portion of its demand curve. Which of the following will occur if the monopolist decreases its price?

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(A) Marginal revenue will decrease, but profits will increase.
(B) Marginal revenue will increase, but profits will decrease.
(C) Total revenue will decrease, but profits will increase.
(D) Both total revenue and profits will decrease. ✓
(E) Both total revenue and profits will increase.

5. Assume that demand for bottled water is relatively price elastic. An increase in supply of bottled water will result in which of the following?

(A) A decrease in price, leading to an increase in total revenue ✓
(B) A decrease in price, leading to a decrease in total revenue
(C) An excess supply of bottled water
(D) An excess demand for bottled water
(E) A relatively small decrease in price and no change in equilibrium quantity

6. Assume that the demand for a certain good is perfectly inelastic and the supply curve of the good is upward sloping. Which of the following occurs in the market for the good if the price of an input used to produce the good increases?

(A) A decrease in both the quantity supplied and the equilibrium amount consumed
(B) A decrease in the quantity supplied and an increase in the equilibrium price
(C) A decrease in the supply and an increase in the equilibrium price ✓
(D) A decrease in both the demand and the equilibrium amount consumed
(E) A decrease in both the quantity demanded and the equilibrium price

7. Assume that the price of good X decreases from $10 to $9 per unit and that the quantity demanded of good X increases from 25 to 30 units. In this price range, the demand for good X is

(A) inelastic
(B) elastic ✓
(C) unit elastic
(D) perfectly inelastic
(E) perfectly elastic

8. If a 10 percent increase in the price of a good leads to a 25 percent decrease in the quantity demanded of the good, demand is

(A) relatively inelastic
(B) relatively elastic ✓
(C) unit elastic
(D) perfectly elastic
(E) perfectly inelastic

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9. Moving from left to right along a downward- sloping linear demand curve, price elasticity varies in which of the following ways?

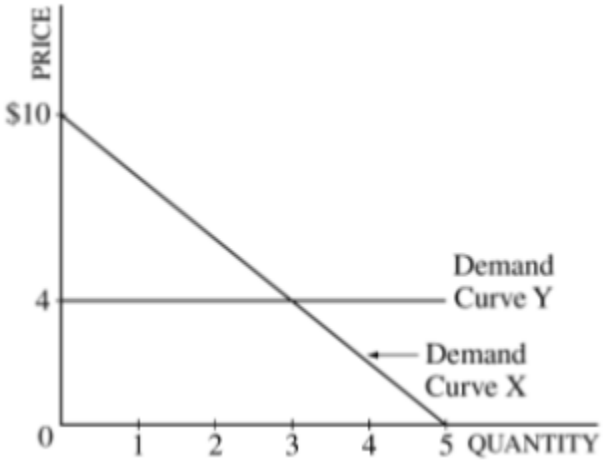
(A) First unit elastic, then inelastic throughout

(B) First unit elastic, then elastic throughout

(C) First inelastic, then unit elastic throughout

(D) First elastic, then unit elastic, and finally inelastic ✓

(E) First inelastic, then unit elastic, and finally elastic



The demand curves X and Y are shown in the graph above.

10. Which of the following offers the most accurate comparison of the price elasticities of demand curves X and Y at a price of $4?

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(A)

Demand Curve X	Demand Curve Y
Perfectly elastic	Perfectly inelastic

(B)

Demand Curve X	Demand Curve Y
Perfectly inelastic	Perfectly elastic

(C)

Demand Curve X	Demand Curve Y
Perfectly inelastic	Relatively elastic

(D) ✓

Demand Curve X	Demand Curve Y
Relatively inelastic	Perfectly elastic

(E)

Demand Curve X	Demand Curve Y
Relatively elastic	Relatively inelastic

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11. Which of the following statements about the price elasticity of demand is true?

(A) When demand is price inelastic, total revenue will decrease as price increases.

(B) When demand is price elastic, an increase in price will increase total revenue.

(C) Demand tends to be more elastic in the short run compared to the long run.

(D) As more close substitutes become available, demand tends to be more price elastic. ✓

(E) As a good becomes viewed as a necessity, demand becomes more price elastic.

Answer D

Correct. With more close substitutes, demand becomes more price elastic because buyers are more

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sensitive to price changes. A slight increase in a good's price will induce the buyers to go for its substitutes.

12. If a severe drought destroys a significant portion of the peanut crop and peanut farmers' revenues increase, which of the following is true over the observed range of prices?

(A) The demand for peanuts must be unit price elastic.

(B) The demand for peanuts must be price elastic.

(C) The demand for peanuts must be unit price inelastic. ✓

(D) The supply of peanuts must be price inelastic.

(E) The supply of peanuts must be price elastic.

13. Which of the following relationships among the price elasticity of demand, change in price, and change in total revenue is consistent?

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(A)

Price Elasticity of Demand	Change in Price	Change in Total Revenue
Elastic	Increase	Increase

(B)

Price Elasticity of Demand	Change in Price	Change in Total Revenue
Elastic	Decrease	Decrease

(C)

Price Elasticity of Demand	Change in Price	Change in Total Revenue
Unit Elastic	Decrease	Decrease

(D)

Price Elasticity of Demand	Change in Price	Change in Total Revenue
Inelastic	Decrease	Increase

(E) ✓

Price Elasticity of Demand	Change in Price	Change in Total Revenue
Inelastic	Decrease	Decrease

Answer E

Correct. When demand is inelastic, the price effect outweighs the quantity effect; that is, the percentage

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change in price exceeds the percentage change in quantity demanded. Thus, a decrease in price will result in a decrease in total revenue.

14. A 10 percent increase in the price of a good results in a 4 percent increase in total revenue. From this information, it can be concluded that the demand over this range of prices

(A) is upward sloping

(B) is inelastic ✓

(C) has a price elasticity of demand equal to 2.5

(D) has increased by 14%

(E) has increased by 40%

Answer B

Correct. The demand elasticity can be determined by applying the total revenue test because whether total revenue increases, decreases, or remains the same when price changes depends on the price elasticity of demand. Price and total revenue move in opposite directions if demand is elastic and move in the same direction if demand is inelastic. In this case, an increase in the price resulted in an increase in total revenue. Thus, demand must be inelastic. That is, the percentage increase in price outweighs the percentage decrease in quantity demanded, resulting in a net increase in total revenue

15. Assume that the price of orange juice increases by 40 percent following a crop failure. If the quantity demanded falls by 10 percent, which of the following is true?

(A) The demand for orange juice is elastic.

(B) The price of grapefruit juice, a substitute good, will fall.

(C) The absolute value of the price elasticity of demand for orange juice is 4.

(D) The absolute value of the price elasticity of demand for orange juice is 0.25. ✓

(E) The absolute value of the price elasticity of demand for orange juice is 10.

Answer D

Correct. The absolute value of the price elasticity of demand is 0.25 and is equal to the percentage change in quantity demanded divided by the percentage change in price, $\frac{10\%}{40\%}$, or 0.25.

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16. The price elasticity of demand for a product is 0.5. If the price of the product increases by 20 percent, which of the following will occur?

(A) The quantity demanded of the good will increase by 10%.

(B) The quantity demanded of the good will increase by 20%.

(C) The quantity demanded of the good will increase by 40%.

(D) The quantity demanded of the good will decrease by 10%. ✓

(E) The quantity demanded of the good will decrease by 40%.

Answer D

Correct. The quantity demanded will decrease by 10 percent. The price elasticity of demand illustrates the negative relationship between price and quantity demanded and is equal to the percentage change in quantity demanded divided by the percentage change in price. Therefore, an increase in price by 20 percent will result in a decrease in quantity demanded by 10 percent for an absolute value of a price elasticity of demand of 0.5.

17. A firm estimates that the absolute value of the price elasticity of demand for its signature sandwich is 2. If the firm increases its sandwich price by 10 percent, what will happen to the quantity demanded?

(A) It will increase by 5 percent

(B) It will increase by 20 percent

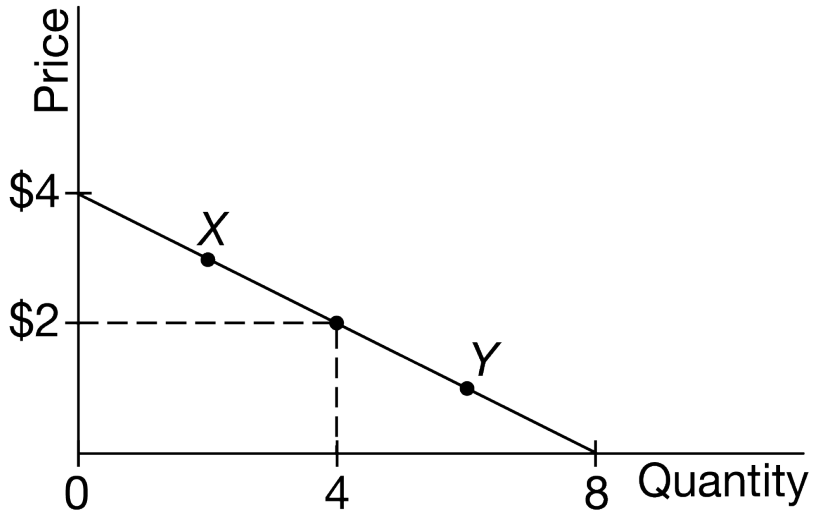
(C) It will decrease by 5 percent

(D) It will decrease by 20 percent ✓

(E) It will remain unchanged

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18.



Which of the following statements is true about the demand curve above?

(A) Demand is elastic at each given price because the slope is constant and equal to -2.

(B) The elasticity of demand decreases when moving from point X to point Y. ✓

(C) The elasticity of demand increases when moving from point X to point Y.

(D) Demand is inelastic between quantities 0 and 4.

(E) Demand is elastic between quantities 4 and 8.

Answer B

Correct. The price elasticity of demand decreases as price falls and quantity rises between point X and Y when moving down along the demand curve. That is, the demand becomes more inelastic along the linear demand curve as price decreases.

19. Which of the following combinations is most likely to result in the demand for a product being highly price inelastic?

(A) The product has few close substitutes and represents a small percentage of a consumer's income. ✓

(B) The product has few close substitutes and represents a large percentage of a consumer's income.

(C) The product has many close substitutes and represents a small percentage of a consumer's income.

(D) The product has many close substitutes and represents a moderate percentage of a consumer's income.

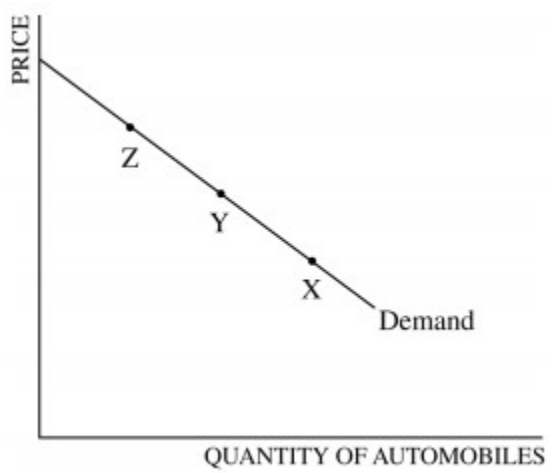
(E) The product has many close substitutes and represents a large percentage of a consumer's income.

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Answer A

Correct. The main determinants of the price elasticity of demand are the number of substitutes available and the percentage of a consumer's income the product represents. The fewer substitutes that are available for a product, all else equal, the more inelastic the demand for the product. Furthermore, the smaller the percentage of a consumer's income a product represents, all else equal, the more inelastic the demand for the product (since an increase in the price, for example, represents a relatively small impact on a consumer's income).

20.



In the figure above, at which of the given points is demand most elastic?

(A) X
(B) Y
(C) Z ✓
(D) The elasticity is the same for all points.
(E) The relative elasticity cannot be determined with the given information.

21. If a 5 percent wage increase in a particular labor market results in a 10 percent decrease in employment, the demand for labor is

(A) perfectly elastic
(B) relatively elastic ✓
(C) unit elastic
(D) relatively inelastic
(E) perfectly inelastic

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22. If a store raises its prices by 20 percent and its total revenue increases by 10 percent, the demand it faces in this price range must be

(A) inelastic ✓

(B) elastic

(C) unit elastic

(D) perfectly elastic

(E) perfectly inelastic

23. If the demand for insulin is price inelastic, a 5 percent increase in the price of insulin will

(A) have no effect on the total revenue of insulin producers

(B) increase the total revenue of insulin producers ✓

(C) decrease the total revenue of insulin producers

(D) decrease the total spending on insulin by consumers

(E) cause the demand for insulin to be less elastic

24. If the demand for product X is perfectly elastic and the supply of product X decreases, which of the following will occur in the market for X ?

(A) Total revenue will increase.

(B) Total revenue will decrease. ✓

(C) Total revenue will remain the same.

(D) Quantity demanded will exceed quantity supplied.

(E) Quantity supplied will exceed quantity demanded.

25. Suppose that the demand for vegetables is price elastic. If the price of vegetables increases by 5 percent, the quantity of vegetables demanded would

(A) increase by 5 percent

(B) increase by more than 5 percent

(C) increase by less than 5 percent

(D) decrease by more than 5 percent ✓

(E) decrease by less than 5 percent

26. To alleviate a financial crisis, a university increases student fees. This action will increase university revenues if the price elasticity of demand for university education is

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(A) inelastic ✓

(B) unit elastic

(C) elastic

(D) equal to the price elasticity of supply

(E) equal to one

27. In which of the following cases would a firm's total revenue increase?

(A) Price increases and demand is elastic.

(B) Price decreases and supply is inelastic.

(C) Price decreases and demand is elastic. ✓

(D) Price decreases and supply is elastic.

(E) Price decreases and demand is inelastic.

28. Which of the following is true in the elastic range of a firm's demand curve?

(A) The firm should expand output to increase economic profits.

(B) An increase in price will also lead to an increase in total cost.

(C) A decrease in price will likely lead to an increase in total revenue. ✓

(D) Marginal revenue is negative.

(E) The firm is maximizing total revenue.

29. Which of the following will tend to make the demand for a product more elastic?

(A) New firms which produce similar products enter the industry. ✓

(B) A change in taste and preferences makes the product more desirable.

(C) The product is necessary for use with a complement.

(D) Production of the product is protected by a patent.

(E) Production cost of the product decreases.

30. The absolute value of the price elasticity of demand for a good increases when

(A) the good has fewer substitutes

(B) the good becomes a necessity

(C) consumers spend greater portion of their budget on the good ✓

(D) the price of an input used to produce the good increases

(E) the good must be purchased immediately

31. The demand curve for shoes is downward sloping, and the absolute value of the price elasticity of demand is 2. If the price of shoes increases by 3 percent, what will happen to the quantity of shoes demanded?

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(A) It will decrease by 6%. ✓

(B) It will decrease by 3%.

(C) It will decrease by 2%.

(D) It will increase by 6%.

(E) It will increase by 3%.

Answer A

Correct. Price elasticity of demand is calculated as the absolute value of the percentage change in quantity demanded/percentage change in price. Therefore, the percentage change in quantity demanded is the product of the price elasticity of demand and the percentage change in price, which is equal to $2 \times 3\% = 6\%$. Since the demand curve for shoes is downward sloping, a 3% increase in the price of shoes will cause the quantity demanded to decrease by 6%.

32. The postmaster general of a country decides to double the price of first-class stamps, expecting the action to cause the revenues from sales of first-class stamps to double. What absolute value of price elasticity of demand is the postmaster general assuming in making the projection?

(A) 0 ✓

(B) Between 0 and 1

(C) 1

(D) Greater than 1

(E) Infinite

Answer A

Correct. For a doubling of the price of first-class stamps to result in a doubling of revenue from stamp sales, the quantity of stamps sold must remain unchanged. When the absolute value of price elasticity of demand is 0, demand is perfectly inelastic indicating quantity demand is not responsive to changes in price. Thus, revenues from stamp sales will increase by the same percentage increase in price, and so doubling of the price will cause total revenue to double.

33. Katie's demand for chocolate bars is perfectly inelastic at a quantity of one per week. Maximiliano will consume three chocolate bars per week when the price is $\$1$, and he will consume four per week when the price is $\$0.50$. Katie and Maximiliano are the only consumers of chocolate bars. Which of the following best describes the price elasticity of market demand beginning at the price of $\$0.50$ and ending at the price of $\$1$?

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(A) Perfectly inelastic

(B) Inelastic ✓

(C) Unit elastic

(D) Elastic

(E) Perfectly elastic

Answer B

Correct. Since Katie and Maximiliano are the only consumers of chocolate bars, the market quantity demanded is 5 when the price is $\$0.50$ and 4 when the price is $\$1.00$. The price elasticity of demand is calculated as the absolute value of the percentage change in quantity demanded/percentage change in price. In this scenario, the percentage change in quantity demanded is $(4-5)/(5) \times 100 = -20\%$. The percentage change in price is $(\$1 - \$0.50)/(\$0.50) \times 100 = 100\%$. This means that the coefficient of the price elasticity of demand coefficient is 0.2 $\left(=\left|\frac{-20\%}{100\%}\right|\right)$, which is in the inelastic range of values for this portion of the demand curve.

34. If the absolute value of the price elasticity of demand for Good X is 0.5, then a 10 percent decrease in the price of Good X will result in which of the following?

(A) A 5% decrease in the quantity demanded of Good X

(B) A 5% increase in the quantity demanded of Good X ✓

(C) A 5% increase in revenues from the sale of Good X

(D) A 10% decrease in revenues from the sale of Good X

(E) A 10% increase in revenues from the sale of Good X

Answer B

Correct. The price elasticity of demand measures the percentage change in quantity demanded resulting from a given change in the price of the good. It also assumes the negative relationship between the price and quantity demanded across the demand curve. When the absolute value of the price elasticity of demand is 0.5, this means that demand is price inelastic. Therefore, the quantity demanded of Good X will increase by 5% $(=0.5 \times 10\%)$.

35. Which of the following must be true if the revenues of wheat farmers increase when the price of wheat increase?

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(A) The supply of wheat is price elastic.

(B) The supply of wheat is income elastic.

(C) The supply of wheat is income inelastic.

(D) The demand for wheat is price elastic.

(E) The demand for wheat is price inelastic. ✓

36. Toby's Toy Trains produces train sets in a monopolistically competitive market. If Toby increases the price of a train set by 5 percent, causing a 10 percent decrease in the quantity demanded of train sets, which of the following will occur?

(A) Toby's total revenue will increase.

(B) Toby's total revenue will decrease. ✓

(C) Toby's total revenue will remain the same.

(D) The demand for Toby's train sets will increase.

(E) The demand for Toby's train sets will decrease.

Answer B

Correct. The price elasticity of demand is given by the ratio of percentage change in quantity demanded to percentage change in price. In this example, the absolute value of the price elasticity of demand is $\frac{10\%}{5\%} = 2$. Over the elastic region of the demand curve (that is, when the absolute value of the elasticity coefficient is greater than absolute value of 1), an increase in price will cause a decrease in Toby's total revenue.